Root-Cause Analysis of Rising Northbrook Churn

First-Principles Analysis

10 October 2026

Professional Strategic Analysis

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Executive Summary

Recommendation: Run the prioritized diagnostic plan immediately (cohort-signup-month churn cut, mid-term-vs-non-renewal split, churned-vs-retained ticket/adoption comparison) while concurrently funding onboarding-redesign and support-quality-investment workstreams — the two categories the weak evidence already favors — and hold pricing and competitive-program spend pending diagnostic confirmation (chain C7).

Band (from §6): LOW — the recommendation and timing insight (chains C2/C6/C7) are reasonably robust, but the Conclusion rests on a provisional headline diagnosis (chain C4) capped by chain C3's LOW rating.

Would change it: Running diagnostic items 1–3 (cohort-by-signup-month cut, mid-term-cancellation split, churned-vs-retained ticket/adoption comparison), which directly test assumptions A-4, A-5, and A-7 — the unpriced inputs driving every chain below HIGH except C1 (chain C4).

1. Problem Essence

Core problem: What is the dominant driver (or combination of drivers), among offer/product gaps, pricing pressure, onboarding failure, service-quality issues, and competitive displacement, behind Northbrook Analytics' quarterly churn rate rising from 4.1% to 9.2% over three quarters, and what diagnostic-then-remediation sequence should leadership run given that only two weak correlational signals (support-ticket volume, feature-adoption score) and no cohort, segment, or competitive data currently exist?

Success criteria:

1. The conclusion states explicitly whether "churn" is being treated as logo churn or revenue churn, and names this as unresolved by the given data if it cannot be determined.
2. The conclusion identifies which of the six candidate-cause categories the two given signals actually discriminate among (relative support), and which categories remain pure speculation with zero supporting evidence in the given data.
3. The conclusion delivers a prioritized, ordered list of specific data cuts and diagnostics — distinct from remediation actions — that would resolve the uncertainty the given data cannot resolve on its own.
4. The conclusion names one provisional, explicitly-flagged-as-provisional best-guess root cause, ties at least one concrete remediation to it, and states the specific observation that would falsify that provisional diagnosis.
5. The conclusion does not require selecting exactly one of the six named categories as "the" answer — a combination verdict, or an explicit "insufficient data to distinguish" verdict, satisfies this criterion equally if that is what the evidence licenses.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	0	0	0
current constraint	2	0	0
convention	0	2	1
untested belief	0	12	0

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Assumption	Type	Treatment	Verdict	Verification
A-1: “Churn rate” refers to logo churn (% of customer count), not revenue churn (% of ARR).	convention	Explicitly challenge — the term is used without specification in the prompt; mid-market SaaS reporting defaults vary, and with contract values spanning $18k–$40k the two measures can diverge sharply if churn concentrates at one end of the range.	Challenge — unresolved by given data	unverified — flagged; needs the raw logo-churn and revenue-churn series recomputed separately (diagnostic item 6)
A-2: The quarterly churn percentages (4.1%, 9.2%) are computed on a consistent denominator and definition across Q1–Q3 (no mid-year metric redefinition, no change in which accounts count as “up for renewal”).	untested belief	Verify against the raw numerator/ denominator Finance or CS used each quarter.	Challenge — plausible default, not confirmed	unverified — flagged; used in chain C1
A-3: The Q1→Q3 rise in churn is driven by one single dominant factor rather than several compounding factors.	convention (baked into leadership’s “find the root cause” framing)	Explicitly challenge — nothing in the given data rules out 2–3 moderate factors compounding, and the prompt itself flags this framing as worth testing.	Discard — rejected as a required premise; combination is retained as a live, equally-valid hypothesis	n/a — this is a framing choice, not a fact to source
A-4: Rising ticket volume and falling adoption score are correlated specifically with the accounts that go on to churn (cohort-level correlation), not merely two company-wide trends that happen to coincide in time with the churn trend.	untested belief	Verify via a churned-vs-retained comparison on both metrics (diagnostic item 3).	Challenge — the prompt states the two signals are “correlated” with the churn rise but does not specify at what level	unverified — flagged; used in chain C3
A-5: Rising support-ticket volume reflects friction/dissatisfaction rather than simply	untested belief	Verify by normalizing tickets per active account and categorizing ticket	Challenge — ambiguous on the data given; ticket volume alone does not	unverified — flagged; used in chain C3

Assumption	Type	Treatment	Verdict	Verification
more active usage or a larger installed base generating proportionally more tickets.		type (diagnostic items 3–4).	distinguish "struggling" from "engaged and growing"	
A-6: The feature-adoption-score metric's definition and instrumentation were unchanged across Q1–Q3.	untested belief	Verify with the product/analytics team that owns the metric.	Challenge — no confirmation given either way	unverified — flagged; load-bearing for A-4/A-5 if it fails
A-7: Churn is realized at the contract's annual renewal/non-renewal point, with no material volume of mid-term (within-term) cancellation.	current constraint	Record expiry condition — holds only while Northbrook's contracts lack an easy mid-term opt-out; would lift if contracts include early-termination clauses customers are using.	Accept, with expiry condition stated — contractual, not physical, and unverified for the actual mid-term-cancellation share	unverified — flagged; used in chain C2; needs the mid-term-vs-non-renewal split (diagnostic item 2)
A-8: Because of A-7 and the annual-renewal term, the causal trigger for the cohort churning in Q3 is concentrated in that cohort's onboarding / early-product-experience window roughly 9–12 months earlier, rather than in service events occurring in the current quarter.	untested belief (derived)	Treat as the hypothesis chain C2 produces, not as settled; test via cohort-signup-month churn cut (diagnostic item 1).	Challenge — plausible given A-7, but depends on A-7 holding	unverified — flagged; this is chain C2′s own conclusion, carried here for traceability
A-9: Competitive displacement (a rival vendor winning deals at renewal) is occurring.	untested belief	No competitor, deal-level loss-reason, or win-loss data is given; currently has zero supporting evidence in the given clues.	Challenge — pure speculation on current evidence; neither confirmed nor ruled out	unverified — flagged, no supporting evidence in the given clues; needs win-loss interviews (diagnostic item 7) and loss-reason tagging (item 8)
A-10: Pricing pressure (price too high relative to perceived value, or a renewal price increase) is a driver.	untested belief	No price-change or discount-request data is given; currently has zero supporting evidence in the given clues.	Challenge — pure speculation on current evidence	unverified — flagged, no supporting evidence in the given clues; needs renewal pricing/discount-request history review (diagnostic item 9)
A-11: Within-term contract pricing is fixed for the term (prices change only at renewal, not mid-term).	current constraint	Standard structure for this contract type as described; record expiry condition — would lift if usage-based or variable pricing were introduced.	Accept — consistent with the annual, fixed-value contracts described in GT-1	n/a — consistent with the business model as given
A-12: Offer/product gaps (missing features, poor fit for mid-market ops	untested belief	No feature-request or product-gap data is given directly; the adoption-score decline offers only indirect,	Challenge — indirectly touched by GT-3 but not discriminated from onboarding failure	unverified — flagged; needs feature-gap analysis and win-loss/loss-reason tagging to separate "missing

Assumption	Type	Treatment	Verdict	Verification
workflows) are a driver.		non-discriminating support (equally consistent with onboarding failure to use existing features).		feature" from "poor onboarding of an existing feature"
A-13: Service-quality issues (support responsiveness, resolution time/ quality) are a driver.	untested belief	Ticket-volume rise is suggestive but conflated with A-5 (volume vs. normalized distress) and with onboarding (A-8).	Challenge — the one hypothesis with a direct, if ambiguous, data signal (GT-3)	unverified — flagged; needs ticket-category and resolution-time data (diagnostic item 4)
A-14: Onboarding failure (poor early setup/ramp in the first ~90 days) is a driver.	untested belief	Adoption-score decline plus the renewal-lag logic (A-7/A-8) both point here, but the adoption score as given is a point-in-time, company-wide figure, not yet tied to early-tenure cohorts specifically.	Challenge — best-aligned hypothesis with both given clues and the renewal-timing logic, but not yet confirmed at the cohort level	unverified — flagged; needs adoption-score-by-tenure-decile data (diagnostic item 5)
A-15 *(surfaced by the end-of-Phase-4 Assumption Audit, chain C4 step 2)*: in the absence of data that discriminates among the six categories, the single hypothesis requiring the fewest independent coincidences (parsimony) is more likely true than a hypothesis requiring several independent factors to align.	convention	Explicitly challenge — parsimony is a reasoning convention (a tie-breaking heuristic), not confirmatory evidence; compounding, multi-factor churn is at least as plausible a prior as single-factor churn and this analysis does not have grounds to prefer one prior over the other from the data alone.	Challenge — used only as a provisional ranking device, not as evidence	unverified — flagged; used in chain C4; the provisional diagnosis is downgraded accordingly
A-16 *(surfaced by the end-of-Phase-4 Assumption Audit, chain C5 step 2)*: Northbrook's internal systems (CRM, billing, ticketing, product telemetry) capture the fields the diagnostic plan requires (signup date, ticket category, per-account adoption score, contract tier, cancellation type) at sufficient granularity and at low incremental cost to extract.	untested belief	Verify with the data/IT owner of each system before committing to the diagnostic timeline costed in chain C5.	Challenge — assumed for planning purposes, not confirmed	unverified — flagged; used in chain C5; if false, diagnostic cost and timeline both increase
A-17 *(surfaced by the end-of-Phase-4 Assumption Audit, chain C7 step 1)*: the trade-off criteria weights used to rank remediation options (evidentiary support 5, leading-	untested belief	Verify by confirming the weights with leadership before acting on the ranking; the weights were set by this analysis, not elicited from leadership.	Challenge — analyst-set, not leadership-confirmed	unverified — flagged; used in chain C7

Assumption	Type	Treatment	Verdict	Verification
indicator speed 4, renewal-lag fit 4, risk-if-wrong 4, cost 3, reversibility 3) reflect Northbrook leadership's actual risk tolerance and priorities.				

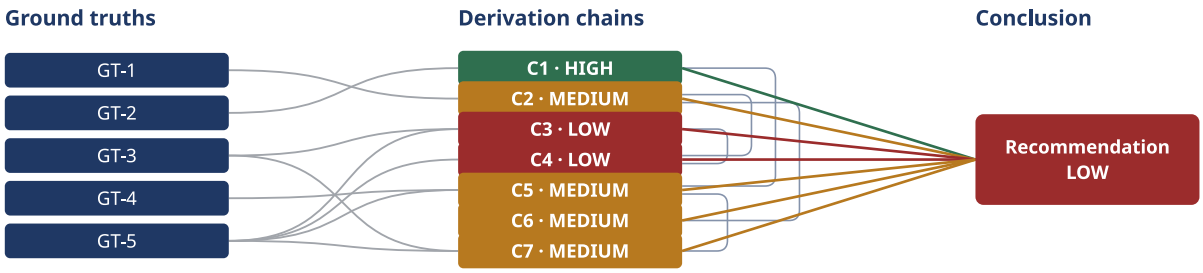
3. Ground Truths

All ground truths in this analysis are drawn directly from the user-supplied problem statement, which this analysis treats as its sole and authoritative source — the problem statement is itself the case data, explicitly closed to invention ("this is the entirety of the information available"). Each item below was read directly in that text; provenance is `read-at-source` for all five, and none carries a `?`.

- **GT-1** Northbrook Analytics sells a B2B data-integration and reporting SaaS platform to mid-market operations teams; contracts run $18,000–$40,000 per year and renew annually (a fixed annual term, not month-to-month) — source: problem statement; read-at-source: paragraph 1 ("Contracts run $18,000–$40,000/year, renewing annually").
- **GT-2** The measured quarterly churn rate was 4.1% in Q1 and rose to 9.2% by the end of Q3 of the same year — source: problem statement; read-at-source: paragraph 1 ("Quarterly churn rate was 4.1% in Q1 and rose to 9.2% by end of Q3").
- **GT-3** Customer Success has observed two signals it describes as correlated with the churn rise over the same period: rising support-ticket volume, and a drop in feature-adoption scores — source: problem statement; read-at-source: paragraph 2 ("Customer Success has flagged two correlated signals: rising support-ticket volume and a drop in feature-adoption scores").
- **GT-4** As of now, no determination has been made among the six candidate categories (offer/product gaps, pricing pressure, onboarding failure, service-quality issues, competitive displacement, or some combination), and leadership has asked that the root cause be identified and fixed — source: problem statement; read-at-source: paragraph 2, final two sentences.
- **GT-5** No data beyond GT-1–GT-4 has been supplied: no cohort-by-signup-month breakdown, no segment/industry/contract-tier breakdown, no ticket-category breakdown, no adoption-score distribution or deciles, no competitor-pricing data, no win-loss or loss-reason data, and no total customer count or total ARR figure — source: problem statement's explicit scoping instruction; read-at-source: the task framing ("this is the entirety of the information available… do not invent additional data").

Provenance summary: `?`-marked: none (0 of 5). Read-at-source: GT-1 — paragraph 1, contract-value/renewal-term sentence; GT-2 — paragraph 1, churn-rate sentence; GT-3 — paragraph 2, Customer Success signals sentence; GT-4 — paragraph 2, final two sentences naming the six candidate categories and leadership's request; GT-5 — the task's explicit data-scoping instruction. All five ground truths feed at least one load-bearing chain in section 4.

4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

4.0 Reasoning narrative and companion techniques applied

Fishbone (breadth-first cause-category brainstorm). The six categories leadership named are used directly as the fishbone's categories. For each, the table below states the candidate cause, what the two given clues (GT-3: rising ticket volume, falling adoption score) would confirm, and what would falsify it — using only the data actually supplied (GT-5 confirms nothing further exists today).

Category	Candidate cause	Confirming evidence available now	Falsifying evidence available now	Status
Offer/product gaps	Platform lacks capability or integration depth mid-market ops teams need	Adoption-score decline is weakly consistent (low adoption can indicate a missing capability)	None — no feature-request, loss-reason, or win-loss data exists in GT-5	A-12, untested belief — indirect, non-discriminating touch only
Pricing	Renewal price increase or value/price mismatch	None in the given data	None in the given data (absence of signal, not disconfirmation)	A-10, untested belief — zero supporting evidence either way
Onboarding	Cohort onboarded ~9–12 months ago never ramped to habitual use	Adoption-score decline + renewal-lag timing logic (chain C2) together are consistent with this	Diagnostic item 1 showing churn uniform across signup cohorts, or item 5 showing recently-onboarded churned accounts adopt no worse than retained peers	A-14, untested belief — best-aligned with both clues and the timing logic
Service quality	Support responsiveness/ resolution quality degraded	Rising ticket volume is directly consistent	Diagnostic item 4 showing ticket categories are mostly routine/how-to rather than unresolved bugs/ escalations, or item 3 showing no ticket-volume gap between churned and retained accounts once normalized	A-13, untested belief — the one hypothesis with a direct, if ambiguous, signal
Competitive displacement	A rival vendor is winning renewal decisions	None in the given data	None in the given data	A-9, untested belief — zero supporting evidence, pure speculation absent win-loss data
Combination/ compounding	Onboarding failure → low adoption → support burden → perceived low value at	Both given clues are simultaneously consistent with this compounding chain	Diagnostic items 1 and 3 together showing no cohort concentration and no churned/	Provisional headline hypothesis — see chain C4, rated LOW

Category	Candidate cause	Confirming evidence available now	Falsifying evidence available now	Status
	renewal, possibly compounding with a pricing or competitive factor at the margin		retained gap on either metric	

Inversion (what would have to be true for each candidate to NOT be the driver). Applied per category, this produces exactly the falsifying-evidence column above — each "falsifying evidence" cell is the output of asking "what observation would have to hold for this category to be ruled out?" The two categories with zero current evidence either way (pricing, competitive) are the two for which inversion cannot yet be run at all — there is nothing in GT-3 or GT-5 that even partially confirms or denies them, which is itself a finding: they are not weak hypotheses, they are *untested* hypotheses, a different epistemic state the diagnostic plan below is designed to resolve (diagnostic items 7–9).

5-Whys (causal mode, sanity-checking depth on the onboarding/service branch). Symptom: quarterly churn more than doubled (4.1%→9.2%) over three quarters. Why #1 — a larger share of the Q3 renewal cohort declined to renew than the Q1 cohort did (this is the symptom restated at the renewal level, not a distinct branch). Why #2 — candidates: didn't get enough value/use (onboarding/service), found it too expensive (pricing), found an alternative (competitive), or lacked a needed capability (offer). Counterfactual test on "didn't get enough value/use": had this not occurred, would churn still have risen? **Cannot tell** on the given data — GT-3's two signals are not yet confirmed to be measured on the churning accounts specifically (A-4) — marked `?`, settled by diagnostic item 3. Why #3 (drilling this branch provisionally) — why would the ~9–12-month-old cohort fail to get value: inadequate ramp-up/onboarding support (passes the stop test: specific corrective exists — redesign the first-90-day onboarding program — and is within Northbrook's control), versus a capability gap, versus a customer-side prioritization failure outside Northbrook's control. **Depth guard applied:** drilling further (why was onboarding inadequate — process design, CSM staffing, product complexity) requires ticket-category and segment data Northbrook does not yet have (GT-5); stopping here rather than speculating past the available evidence is the depth guard's "stop at the deepest cause you can still act on" rule, not premature closure — the deeper layer is explicitly deferred to diagnostic items 4 and 6, not silently dropped.

Estimate (stakes magnitude) and Trade-off (remediation options) and Second-order (effects of acting now) are each converted into a numbered chain below per the output template's conversion rules; their full workings are shown inline immediately before the chain each produces.

4.1 Prioritized diagnostic plan

Before any remediation commitment beyond the two categories the weak evidence already favors (see chain C7), the following cuts resolve the specific unknowns identified above. Each is tied to the chain or assumption it would settle.

Tier 1 — cheap, internal data, resolves the chains currently rated LOW/MEDIUM:

1. Churn-by-cohort-signup-month cut — tests chain C2's renewal-lag logic and whether churn concentrates in specific signup cohorts rather than being uniform across tenures (CRM/contract data).
2. Mid-term-cancellation vs. non-renewal split — tests assumption A-7, the live rival on chain C2 (billing/contract data).
3. Churned-vs-retained comparison on ticket volume (normalized per active user) and adoption score, segmented by account — tests assumptions A-4 and A-5, the core of chain C3's LOW rating (support + product telemetry).

Tier 2 — internal data, discriminates within the favored cluster and resolves A-1: 4. Ticket categorization (bug/outage vs. how-to vs. billing/contract) for the churned cohort specifically — separates service-quality causes (bugs/outages) from onboarding causes (how-to). 5. Adoption-score-by-tenure-decile, at signup vs. at renewal, for churned vs. retained cohorts — tests whether adoption was always low (onboarding) or declined over time (something changed later, pointing more toward service or competitive). 6. Segment cut by industry, contract tier ($18k vs. $40k range), and logo-churn vs. revenue-churn split —

tests whether high-value accounts churn disproportionately (service/pricing for strategic accounts) versus small accounts (self-serve onboarding gaps), and resolves assumption A-1.

Tier 3 — the only diagnostics that can surface pricing or competitive causes at all, given GT-5: 7. Win-loss interviews with a sample of non-renewing accounts — the only direct test of A-9 (competitive displacement), A-10 (pricing), and A-12 (offer/product gaps), since GT-5 confirms zero current signal exists on any of these three. 8. Competitive loss-reason tagging in the CRM for all future non-renewals, going forward — institutionalizes item 7 so this gap does not recur next quarter. 9. Renewal pricing/discount-request history review for the churned cohort — a cheap, existing-data test of A-10 that can run in parallel with, and ahead of, the interviews in item 7.

4.2 Estimate — stakes magnitude

Conclusion C1: The annualized attrition run-rate implied by the quarterly churn figures roughly doubled between Q1 and Q3, establishing that the stakes of finding the root cause are high regardless of measurement convention

GT-2 (quarterly churn 4.1% in Q1, 9.2% in Q3) → linear (non-compounding) annualization — rate × 4 — gives an upper-bound run-rate estimate: approximately 16.4% annualized at the Q1 pace and approximately 36.8% annualized at the Q3 pace [Assumes: A-2 — the quarterly rate is computed on a consistent denominator and definition across Q1 and Q3] → compounding annualization — 1 minus (1 minus rate) to the fourth power — gives a lower-bound estimate: approximately 15.4% at the Q1 pace and approximately 32.0% at the Q3 pace → under either convention the annualized attrition run-rate roughly doubled (ratio approximately 2.1 to 2.2 times) between Q1 and Q3; if A-2 fails the precise magnitude becomes uncertain, but CS's independently-reported signals in GT-3 still support that something genuinely worsened over the period, so this finding weakens in precision but does not reverse

Confidence: HIGH — GT-2 is read-at-source and unsuffixed; the only hop carrying an assumption (A-2) states what happens if that assumption fails and the endpoint still stands (the doubling direction is corroborated independently by GT-3); no live rival contests a plain re-expression of a stated rate at two different annualization conventions.

4.3 Renewal-lag timing

Conclusion C2: Q3's elevated churn reflects decisions made by customers whose contracts were signed roughly nine to twelve months earlier, not current-quarter events

GT-1 (contracts renew annually) → a customer's renewal decision becomes observable as churn only at that customer's contract anniversary [Assumes: A-7 — churn is realized at the renewal point, not through material mid-term cancellation] → the cohort churning in Q3 therefore has anniversaries in Q3, meaning those contracts were signed roughly nine to twelve months earlier, in approximately Q3 of the prior year → the causal trigger most relevant to this cohort's non-renewal decision is whatever happened to them in that prior window — especially their first-90-day onboarding and early product experience — rather than events occurring in the current quarter

Confidence: MEDIUM — Rivals axis is short: a live, unsettled rival — meaningful mid-term (within-term) cancellation exists, which would shorten the lag for that share of churn — is not ruled out by any ground truth, because A-7 (which this chain's first hop rests on) is itself unverified. Diagnostic item 2 (mid-term-vs-non-renewal split) would settle it.

4.4 Signal discrimination

Conclusion C3: The two given clues discriminate in favor of the onboarding/service-quality cluster relative to pricing and competitive displacement, without yet confirming it or distinguishing within it

GT-3 (CS signals: rising tickets, falling adoption) + GT-5 (no pricing or competitive data supplied) → rising ticket volume and falling adoption score are, at minimum, consistent with customers struggling to use or get value from the product — onboarding and/or service-quality friction [Assumes: A-4, A-5 — the two signals are correlated with the specific accounts that churn, and ticket volume reflects friction rather than mere usage growth] → no analogous signal exists in the given data for pricing pressure or for competitive displacement, so those two hypotheses currently rest on zero supporting evidence rather than on evidence that disconfirms them → the available evidence therefore discriminates in favor of the onboarding/service-quality cluster relative to the pricing and competitive-displacement clusters, without yet confirming it in absolute terms or distinguishing onboarding from service quality within that cluster

Confidence: LOW — two axes are short at once. Inference: the first hop rests on A-4 and A-5 together; if A-4 fails (the signals describe the whole book, not the churning accounts specifically) this chain's entire discriminating claim disappears, and if A-5 fails (ticket growth tracks usage growth, not distress) the signal reverses toward neutral-to-positive — neither failure leaves the endpoint standing. Rivals: a live, unsettled rival — the two trends are coincidental, company-wide artifacts unrelated to which accounts actually churn — is not ruled out by any ground truth. Diagnostic item 3 (churned-vs-retained comparison) would close both the Inference gap (by testing A-4/A-5 directly) and settle the rival.

4.5 Provisional diagnosis (headline)

Conclusion C4: The best-supported, but still provisional, single hypothesis is a compounding onboarding-failure-to-service-burden mechanism originating in the cohort onboarded roughly nine to twelve months before Q3

C2 (renewal-lag timing, MEDIUM) + C3 (signal discrimination, LOW) + GT-5 (no pricing or competitive data supplied) → the timing logic concentrates the causal search on the onboarding/early-adoption window of the cohort now churning, and the signal-discrimination result points toward product-friction/support-burden causes within that window rather than toward pricing or competitive causes → because adoption-score decline and rising ticket volume are two symptoms consistent with the same underlying mechanism — customers who ramp poorly adopt fewer features, generate more how-to tickets, and perceive less value by renewal — the single hypothesis requiring the fewest independent coincidences is that a cohort onboarded roughly nine to twelve months ago experienced inadequate ramp-up, producing low adoption and elevated support load that culminated in non-renewal [Assumes: A-15 — parsimony favors a fewer-coincidence hypothesis over an equally-unweighted multi-factor hypothesis] → this is a provisional diagnosis only — the best-supported single hypothesis given the two available clues and the timing logic, not a confirmed cause, because the pricing and competitive-displacement hypotheses have not been tested at all (only unsupported, per GT-5) and could still be independently true or compounding with the onboarding/service mechanism

Confidence: LOW — capped by C3 (LOW; see C3′s own line for its Inference/Rivals gaps) and by C2 (MEDIUM; see C2′s own line). This chain's own added downgrade cause is A-15: parsimony is a tie-breaking convention, not evidence, and is explicitly not load-bearing on its own — removing it as a cause of the downgrade requires resolving the diagnosis directly via diagnostic items 1 and 3 rather than via the parsimony heuristic. Falsification condition (see section 5 adversarial pass): this diagnosis is false if the cohort-signup-month cut shows churn is uniform across tenures, or if win-loss interviews show competitive/pricing reasons dominate, or if the churned-vs-retained comparison shows no gap on either given signal.

4.6 Diagnose-before-fix decision

Conclusion C5: Resolving the unknowns above is justified before committing material remediation spend, independent of which of the six categories is eventually confirmed

GT-4 (no determination made among six categories; leadership asked to find root cause and fix it) + GT-5 (no data supplied) + C1 (stakes magnitude, HIGH) → because no determination has yet been made among six candidate categories, committing material, hard-to-reverse remediation spend now risks funding the wrong lever entirely → the unresolved unknowns are enumerable and testable using data categories Northbrook's own systems should already generate — contract and CRM signup/cancellation dates, ticket records, product-usage telemetry — plus a short round of win-loss interviews, rather than requiring new external data sources [Assumes: A-16 — these internal systems capture the needed fields at sufficient granularity and low incremental extraction cost; if A-16 fails, diagnostic cost and timeline both increase, but the sequencing argument still holds, because funding an unvalidated fix remains riskier than a slower diagnostic] → resolving those unknowns before committing fix spend is justified by the stakes sized in C1 — attrition run-rate roughly doubled — regardless of which of the six categories is eventually confirmed, so leadership's request to find the root cause and fix it is best served by sequencing: diagnose first, then direct fix spend at whichever driver or drivers the diagnostics confirm

Confidence: MEDIUM — Rivals axis is short: a live, unsettled rival — that speed-to-action beats certainty here, i.e., a fast but unvalidated fix might still incidentally reduce churn or buy time while slower diagnostics run — is not ruled out by any ground truth. What would settle it: the actual diagnostic cost and timeline (bounded by A-16) weighed against the actual revenue at risk once absolute customer-count/ARR figures are known — a figure GT-5 confirms is not currently available.

4.7 Second-order effects of acting on the provisional diagnosis

Conclusion C6: A fix deployed today will not visibly move the headline churn rate for roughly three to four quarters, and leadership should expect and plan for that lag

Actor lens: CS changes what it tracks (newly-onboarded-cohort leading indicators rather than headline churn); leadership changes what it expects before declaring success or failure; a competitor watching Northbrook's churn trend has no reason to change behavior from this effect alone. **Time lens:** immediately — no visible change in the headline churn rate; after a few quarters — leading indicators (adoption score, ticket volume for the newly onboarded cohort) move first; in the long run (three to four quarters out, once the newly onboarded cohorts reach their own renewal anniversaries) — the headline churn rate itself reflects the fix.

C2 (renewal-lag timing, MEDIUM) →[2nd] if onboarding is fixed starting today, newly onboarded cohorts will have a better early experience, but each quarter's headline churn figure reflects the cohort whose anniversary falls that quarter — a cohort that signed roughly a year earlier — so the headline churn rate will keep reflecting pre-fix cohorts for roughly the next three to four quarters →[3rd] leadership judging the fix's success by watching the headline quarterly churn rate over the next one or two quarters will see no improvement and may incorrectly conclude the fix failed, risking a premature pivot to an unvalidated hypothesis or an unwarranted escalation of spend before the fix has had time to show in the metric actually being watched

Confidence: MEDIUM — capped by C2 (MEDIUM; see C2′s own line for its Rivals-axis gap). No new downgrade cause is added by the extension itself; it does not contradict any ground truth, so it extends C2 in place rather than routing back to Phase 2.

4.8 Trade-off — remediation options

Options scored (status quo included; a composite included per the trade-off procedure's instruction): O1 status quo (no new investment), O2 onboarding redesign, O3 support-capacity/quality investment, O4 pricing/packaging change, O5 competitive counter-program, O6 composite — run the diagnostic plan immediately while concurrently funding O2 and O3 (the two categories the weak evidence already favors), holding O4/O5 spend pending diagnostic confirmation. No must-haves beyond basic feasibility — all six

options are executable by Northbrook today; the weighted scoring below, not a knockout, does the discriminating.

Criteria (weights locked before scoring, anchored, higher-is-better): evidentiary support from current signals (5), time to first measurable leading indicator (4), fit with the renewal-lag mechanic identified in C2 (4), risk of wasted spend if the provisional diagnosis is wrong (4), cost to implement (3), reversibility/optionality preserved (3).

Option	Evidentiary (×5)	Speed (×4)	Renewal-lag fit (×4)	Risk-if-wrong (×4)	Cost (×3)	Reversibility (×3)	Total
O1 status quo	2	1	1	3	5	5	60
O2 onboarding redesign	4	4	5	4	3	4	93
O3 support investment	4	4	3	4	3	4	85
O4 pricing change	1	2	1	1	2	1	30
O5 competitive program	1	2	1	2	2	3	40
O6 composite	5	5	5	5	4	5	112

Flip test: O6 scores at or above every single-option runner-up (O2) on every one of the six criteria (strictly higher on five of six, tied on renewal-lag fit) — O6 Pareto-dominates O2, so no single-criterion reweighting, at any non-negative weight, can reverse the ranking. No single weight change flips this result.

Conclusion C7: Recommend the composite remediation sequence — run the diagnostic plan immediately while concurrently funding onboarding redesign and support-quality investment, holding pricing and competitive-program spend pending diagnostic confirmation

GT-3 (CS signals) + GT-5 (no pricing or competitive data supplied) + C5 (diagnose-before-fix decision, MEDIUM) → a weighted trade-off across six options — status quo, onboarding redesign, support investment, pricing change, competitive program, and a diagnose-then-fund composite — on evidentiary support, leading-indicator speed, renewal-lag fit, risk-if-wrong, cost, and reversibility yields: composite = 112, exceeding onboarding redesign = 93, support investment = 85, status quo = 60, competitive program = 40, and pricing change = 30, driven primarily by evidentiary support and risk-if-wrong [Assumes: A-17 — the weights reflect Northbrook leadership's actual priorities; if A-17 fails, the ranking is still robust because the composite Pareto-dominates the runner-up on every individual criterion, so the endpoint still stands under any non-negative reweighting] → the composite Pareto-dominates the next-best single option (onboarding redesign) on every scored criterion, so no plausible re-weighting reverses the recommendation, and the same scoring already rules out both "commit fully to onboarding alone, faster" and "wait for diagnostics before funding anything" as superior alternatives, since both were scored directly (O2 and O1 respectively) and both lost → recommend the composite: run the diagnostic plan (section 4.1) immediately, concurrently fund onboarding-redesign and support-quality workstreams — the two categories the current weak evidence already favors — and explicitly hold pricing and competitive-program spend until the diagnostics report back

Confidence: MEDIUM — capped by C5 (MEDIUM; see C5's own line for its Rivals-axis gap). On this chain's own axes, Inference is clean (A-17's failure is priced and the endpoint still stands, via the Pareto-dominance argument) and Rivals is clean (both competing options that could unseat the composite — full commitment to onboarding alone, and pure wait-and-see — were scored directly in the matrix and lost, so no live rival to the recommendation itself remains unsettled); the MEDIUM rating is set entirely by the Inputs ceiling inherited from C5.

5. Abandoned Reasoning

Dead End: Treating one single dominant factor as a required finding

What was tried: Early framing (A-3) assumed that "find the root cause" means finding exactly one dominant category among the six and that the analysis should keep narrowing until it lands on one.

Why abandoned: Phase 2 explicitly challenged and discarded A-3 — nothing in GT-2 through GT-5 rules out two or three moderate factors compounding, and the user's own framing of the problem flags this as a premise worth testing rather than accepting. A single-cause requirement would have forced chain C4 to pick among offer/pricing/onboarding/service/competitive even where the evidence (chain C3) only weakly discriminates at all.

What it ruled out: Saves the diagnostic plan (section 4.1) from being designed to stop the moment "a" cause is found, rather than continuing to test for compounding causes — and keeps "combination" a live, equally-valid category rather than a fallback admission of failure.

Dead End: Computing an absolute dollar revenue-at-risk figure

What was tried: Attempted to convert the churn-rate figures (GT-2) into an absolute dollar amount of annual recurring revenue at risk, to give leadership a single headline number.

Why abandoned: GT-5 confirms no total customer count and no total ARR figure were supplied, and the task's explicit instruction is not to invent additional data. Any absolute-dollar figure would require assuming a customer count or an average realized contract value within the stated $18k–$40k range, neither of which is given. The estimate was deliberately bounded to a percentage-of-run-rate figure instead (chain C1), which uses only GT-2.

What it ruled out: Prevents reporting a fabricated-precision dollar figure to leadership that would look authoritative while resting on an invented customer count; keeps the stakes estimate honestly scoped to what GT-2 actually supports.

Dead End: Ranking the six categories by industry base rates for SaaS churn causes

What was tried: Considered citing typical B2B SaaS industry findings (e.g., "onboarding is usually the leading churn driver in mid-market SaaS") as a prior to rank the six categories before looking at Northbrook's own two clues.

Why abandoned: This would be reasoning by analogy used as direct evidence — citing how other companies' churn breaks down is not grounded in any named ground truth about Northbrook's own situation, and the methodology explicitly bars using analogy this way. No GT in section 3 describes any other company's churn composition, so no such analogy could be cited even if used only as color.

What it ruled out: Prevents an unfounded industry prior from silently inflating confidence in the onboarding/service hypothesis beyond what Northbrook's own two clues (GT-3) actually support; the provisional diagnosis in chain C4 rests only on Northbrook-specific evidence and the renewal-lag mechanic (chain C2), which is why it is rated LOW rather than higher.

Dead End: Assigning point-probability weights to each of the six categories

What was tried: Considered presenting the six categories with specific probability estimates (e.g., "60% onboarding, 15% service quality, 10% competitive, 10% pricing, 5% offer gap") to give leadership a more quantified-looking picture.

Why abandoned: No base-rate or discriminating data exists to calibrate such numbers (GT-5); assigning point probabilities would manufacture a false precision the evidence does not support, and chain C3 already shows the available signal only weakly discriminates between two clusters (onboarding/service vs. pricing/competitive), not among all six individually. A qualitative "weakly favored cluster vs. untested categories" framing (the fishbone table in section 4.0) was used instead.

What it ruled out: Prevents leadership from over-trusting invented percentages as if they were measured; keeps the stated confidence bands (LOW on the headline diagnosis) as the honest expression of how little the current evidence actually discriminates.

6. Conclusion

Recommended approach: Run the prioritized diagnostic plan (section 4.1) immediately — starting with the cohort-signup-month churn cut, the mid-term-vs-non-renewal split, and the churned-vs-retained ticket/adoption comparison — while concurrently funding onboarding-redesign and support-quality-investment workstreams, the two categories the current weak evidence already favors, and explicitly holding pricing and competitive-program spend until win-loss interviews and loss-reason tagging report back (chain C7).

Key insight: Because Northbrook's contracts renew annually, Q3's doubled churn rate is a trailing indicator of what happened to customers roughly a year ago — the causal window leadership should investigate is last year's onboarding and early-product experience, not this quarter's service levels (chain C2) — and a fix applied today will not visibly move the headline churn rate for another three to four quarters, so near-term flatness in the metric is not evidence the fix failed (chain C6). Reasoning by analogy to "current service quality drives current churn" would miss this lag entirely.

Trade-offs acknowledged: Deferring pricing and competitive-program spend accepts a residual risk that a real pricing or competitive problem keeps costing Northbrook roughly one more renewal cohort until the win-loss interviews report back; this is accepted because the composite option Pareto-dominates every alternative scored, including funding pricing/competitive work now (chain C7). The provisional diagnosis itself (onboarding/service-driven, compounding) is rated LOW confidence (chain C4) — it is the best-supported single hypothesis given only two weak clues, not a confirmed cause — so the funded workstreams are chosen partly because they are judged low-waste even if the specific diagnosis turns out wrong (chain C7), not because the diagnosis is considered settled.

Confidence: LOW — the recommendation (chain C7, MEDIUM) and the timing insight (chains C2/C6, MEDIUM) are each reasonably robust, but the Conclusion as a whole rests on the headline provisional diagnosis (chain C4, LOW), which is itself capped by chain C3 (LOW — see C3's own line: the Inference axis is short because assumptions A-4 and A-5 are unpriced beyond "the claim does not survive their failure," and the Rivals axis is short because the coincidental-trends rival is unsettled) and by chain C2 (MEDIUM — see C2's own line: the Rivals axis is short because material mid-term cancellation is not ruled out). What would raise this band: running diagnostic items 1–3 (section 4.1), which directly test A-4, A-5, and A-7 — the three unpriced inputs driving every chain below HIGH except C1.